Statistical Anomaly Detection System for Fraudulent Banking Transactions: A Comparative Study of Statistical Methods, Machine Learning, and Deep Learning

1. Project Summary

This project proposes the development and evaluation of a statistical anomaly detection system for identifying potentially fraudulent credit card transactions. The study will investigate whether traditional statistical techniques, when combined with machine learning and deep learning methods, can effectively detect fraudulent transactions in a highly imbalanced dataset.

The proposed framework will use Z-score analysis and the Interquartile Range (IQR) as statistical baselines. These methods will be compared with Isolation Forest, an unsupervised machine learning algorithm, and an autoencoder-based deep learning model. A supervised machine learning model will also be included to establish a benchmark using the available fraud labels.

The research is motivated by the limitations of conventional statistical techniques, which generally examine individual features and may fail to identify complex relationships among transaction variables. Recent research has explored deep learning, generative models, and imbalance-aware learning to improve fraud detection (Albalawi & Dardouri, 2025; Ge et al., 2025; Alharbi et al., 2026).

The project will use the publicly available European credit card transaction dataset from Kaggle, containing 284,807 transactions, including 492 fraudulent transactions. The proposed system will be evaluated using precision, recall, F1-score, precision-recall area under the curve (PR-AUC), and false-positive rates.

The principal contribution will be a systematic comparison of statistical, machine learning, and deep learning approaches under the same experimental conditions. The study will also investigate whether combining statistical anomaly scores with machine learning predictions improves fraud detection while keeping false alerts manageable.

2. Background and Motivation

Electronic banking has become an essential component of modern financial services. Credit card payments, online purchases, ATM withdrawals, and electronic money transfers enable customers to conduct financial transactions quickly and conveniently. However, these services also expose customers and financial institutions to fraudulent activities.

Fraud detection is challenging because fraudulent transactions typically represent a very small proportion of all transactions. In the selected Kaggle dataset, only approximately 0.172% of transactions are fraudulent. A model that classifies every transaction as

legitimate would achieve approximately 99.83% accuracy while failing to identify any fraud. Consequently, accuracy alone is not a reliable measure of fraud detection performance.

Traditional fraud detection systems frequently use predefined rules or supervised classification models trained on historical fraud examples. Although these approaches can be effective, they may struggle to identify previously unseen fraud patterns or unusual transactions that do not resemble known fraudulent cases.

Statistical anomaly detection provides an alternative by identifying transactions that deviate from expected patterns. Z-score analysis identifies observations that are unusually far from a feature's mean, while IQR identifies observations outside distribution-based boundaries. These techniques are relatively straightforward to implement and interpret, making them suitable baselines for a fraud detection framework.

However, individual statistical measures have important limitations. Transaction amounts may be highly skewed, making mean- and standard-deviation-based thresholds less reliable. Furthermore, a transaction can be unusual in several features simultaneously without being an outlier in any individual feature. Conversely, a statistically unusual transaction may be legitimate.

Machine learning and deep learning offer ways to address these limitations by learning relationships among multiple features. Isolation Forest isolates unusual observations through random partitioning, while autoencoders learn representations of data and can identify transactions that are poorly reconstructed by a model trained on legitimate transactions.

Recent research demonstrates the growing interest in imbalance-aware and deep learning approaches. Albalawi and Dardouri (2025) investigated conventional machine learning alongside a deep learning model using focal loss. Ge et al. (2025) proposed a gated-attention generative adversarial network for imbalanced credit card fraud detection. Alharbi et al. (2026) examined a framework combining autoencoders, generative augmentation, and ensemble learning.

Despite these developments, more complex models may require greater computational resources and can be more difficult to interpret. Their advantages should therefore be demonstrated empirically rather than assumed.

2.1 Research Motivation

The project is motivated by four principal concerns:

1. **Extreme class imbalance:** Fraudulent transactions are rare, making it difficult to train and evaluate reliable detection systems.

2. **Limitations of individual statistical methods:** Z-score and IQR may fail to detect fraud that depends on interactions among multiple transaction features.
3. **Need for measurable comparisons:** Statistical, machine learning, and deep learning techniques should be evaluated using consistent metrics and data splits.
4. **Practical usefulness:** A fraud detection system should identify a meaningful proportion of fraudulent transactions without overwhelming analysts with false alerts.

The motivation is therefore not simply to build another fraud classifier. It is to investigate whether relatively simple statistical techniques remain competitive with more complex AI methods and whether a hybrid approach offers measurable benefits.

3. Research Aim and Objectives

3.1 Research Aim

The primary aim of this research is to develop and evaluate a comparative anomaly detection framework for fraudulent credit card transactions, investigating the effectiveness of statistical techniques, machine learning, deep learning, and their potential combination under extreme class imbalance.

3.2 Research Objectives

The project will pursue the following objectives:

1. Examine the distribution of legitimate and fraudulent transactions through exploratory data analysis.
2. Implement Z-score and IQR methods to identify statistically unusual transactions.
3. Develop an Isolation Forest model and an autoencoder-based anomaly detection model to capture multivariate deviations from normal transaction patterns.
4. Implement a supervised machine learning benchmark using the available fraud labels.
5. Investigate a hybrid detection strategy that combines statistical anomaly scores with machine learning predictions.
6. Evaluate the methods using precision, recall, F1-score, PR-AUC, and false-positive rates, with particular attention to the minority fraud class.
7. Investigate how detection thresholds affect the trade-off between fraud detection and false alerts.

8. Compare the predictive performance, interpretability, and computational cost of the proposed methods.
9. Identify the limitations of the selected dataset and recommend improvements for future research.

4. Research Questions

The study will address the following research questions.

RQ1: How effectively can Z-score and IQR detect fraudulent transactions in a highly imbalanced credit card dataset?

RQ2: Do multivariate anomaly detection methods, specifically Isolation Forest and an autoencoder, outperform the individual statistical methods in detecting fraudulent transactions?

RQ3: Does combining statistical anomaly scores with machine learning predictions improve fraud detection performance compared with using each method independently?

RQ4: How does the choice of detection threshold affect precision, recall, PR-AUC, and the number of false-positive alerts?

RQ5: What trade-offs exist between detection performance, interpretability, and computational cost across statistical, machine learning, and deep learning approaches?

4.1 Research Hypotheses

The following hypotheses will guide the quantitative evaluation:

- **H1:** Multivariate anomaly detection methods will achieve higher fraud recall than at least one of the individual statistical baselines.
- **H2:** A hybrid statistical and machine learning approach will improve the precision-recall trade-off compared with the best individual statistical method.
- **H3:** Deep learning will not necessarily outperform simpler models on every evaluation metric, particularly when training data are limited and fraud examples are scarce.

The hypotheses will be assessed using experimental results rather than assumed to be true. Where appropriate, repeated experiments and statistical comparisons will be used to determine whether observed differences are meaningful.

5. Literature Review and Research Gap

5.1 Statistical Anomaly Detection

Z-score analysis and IQR are established statistical techniques for identifying observations that differ from a feature's expected distribution. Their principal advantages are simplicity, low computational cost, and interpretability.

However, conventional univariate methods examine features separately. This can limit their ability to detect transactions that are unusual because of combinations of features rather than extreme values in any one feature.

For example, a transaction amount may be common across the dataset, but the transaction's combination of anonymized features may be unusual. A univariate threshold based only on amount may not identify it.

5.2 Machine Learning for Fraud Detection

Isolation Forest is an unsupervised anomaly detection method that identifies observations that can be isolated relatively quickly through random feature selection and partitioning. It provides a multivariate alternative to individual-feature statistical thresholds.

Supervised machine learning models, such as Random Forest, can learn patterns associated with known fraud labels. However, their effectiveness depends on the representativeness of historical fraud examples and appropriate handling of class imbalance.

Ahmed et al. (2025) investigated an ensemble fraud detection approach using hybrid data sampling, including SMOTE and Edited Nearest Neighbours. Their work illustrates the continuing importance of imbalance handling in fraud detection.

5.3 Deep Learning and Recent Developments

Autoencoders are neural networks trained to reconstruct their inputs. When trained primarily on legitimate transactions, they can potentially assign higher reconstruction errors to transactions that differ from the learned normal patterns.

More advanced methods have also been investigated. Ge et al. (2025) proposed a gated-attention generative adversarial network to address imbalanced fraud detection. Albalawi and Dardouri (2025) evaluated traditional machine learning and deep learning approaches, including focal loss. Alharbi et al. (2026) investigated a multistage approach involving autoencoder-based representations and generative ensemble learning.

These studies demonstrate that recent research extends beyond simple statistical thresholds. However, sophisticated models may be more expensive to train, require careful tuning, and depend on sufficient representative training data.

ImDiffusion, introduced by Chen et al. (2024), is another relevant development in multivariate time-series anomaly detection. It combines imputation and diffusion models to capture temporal and inter-feature dependencies. However, it was developed as a general time-series anomaly detection framework, not specifically as a Z-score or IQR-based credit card fraud detector. Its relevance is therefore primarily conceptual unless the project is expanded to include a suitable temporal modelling experiment.

5.4 Identified Research Gap

The reviewed research demonstrates the potential of machine learning and deep learning for imbalanced fraud detection. Nevertheless, a practical question remains: how much additional detection value do these more complex methods provide over simple, interpretable statistical techniques when evaluated consistently?

This project will address that question through a controlled comparison of statistical baselines, multivariate anomaly detection, supervised classification, and a hybrid approach.

The intended contribution is not to claim a new deep learning architecture. Instead, it is to provide a transparent, reproducible evaluation of the relative benefits and limitations of the methods, with particular emphasis on severe class imbalance, false alerts, and computational practicality.

6. Proposed Methodology

6.1 Research Design

The project will adopt a quantitative experimental research design. A single publicly available dataset will be used to implement, test, and compare multiple fraud detection approaches.

The research will follow these stages:

1. Dataset acquisition and inspection.
2. Data preprocessing and exploratory analysis.
3. Implementation of statistical anomaly detection baselines.
4. Development of machine learning and deep learning models.
5. Development of a hybrid anomaly detection strategy.
6. Model evaluation and comparative analysis.
7. Interpretation of results and documentation of limitations.

All experiments will be implemented in Python using Jupyter Notebook.

6.2 Dataset Description

The primary dataset will be the Credit Card Fraud Detection dataset available through Kaggle:

https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

The dataset contains transactions made by European cardholders over two days in September 2013.

Dataset characteristic	Description
Total transactions	284,807
Legitimate transactions	284,315
Fraudulent transactions	492
Fraud prevalence	Approximately 0.172%
Time	Seconds elapsed between transactions and the first transaction
V1–V28	Anonymized principal-component features
Amount	Transaction amount
Class	Target label: 0 = legitimate, 1 = fraud

The dataset is suitable for this project because it contains labelled fraud examples while preserving a highly imbalanced class distribution. It also allows unsupervised anomaly detection methods to be evaluated against known fraud labels without requiring the labels to be used during their training.

However, the dataset has limitations. It covers only two days, its original transaction characteristics are mostly anonymized, and it does not provide the full transaction histories or customer context available in operational banking systems. Consequently, results will be interpreted as benchmark findings rather than proof of real-world banking effectiveness.

6.3 Data Preprocessing

The dataset will be inspected for missing values, duplicate records, inconsistent values, and unexpected data types.

The preprocessing stage will include:

- Loading the dataset using Pandas.
- Examining the number of transactions and distribution of the target label.
- Inspecting the ranges and distributions of numerical features.
- Applying appropriate scaling where required by the selected models.
- Separating input features from the fraud label.
- Establishing training, validation, and test datasets.
- Preventing information leakage between the training and evaluation stages.

The test set will remain untouched until the final evaluation. Threshold selection and model tuning will use training and validation data only.

The primary experiment will use a stratified split to preserve the minority-class proportion. A chronological evaluation will also be considered because the data contain a time feature, although the dataset's short observation period limits the strength of conclusions about future fraud patterns.

Any resampling technique, such as SMOTE, will be applied only to the training data of the relevant supervised model. It will not be applied to validation or test data. For unsupervised models, the fraud labels will be withheld during training so that their detection ability can be assessed independently.

6.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) will be conducted to understand the characteristics of the dataset and the differences between legitimate and fraudulent transactions.

The analysis will include:

1. A bar chart showing the number of legitimate and fraudulent transactions.
2. Histograms and boxplots of transaction amounts.
3. Distribution plots for selected anonymized features.
4. Comparison of selected feature distributions between legitimate and fraudulent transactions.
5. Correlation analysis among numerical features.
6. Analysis of the relationship between transaction amount and fraud labels.

7. Investigation of how statistical outlier thresholds affect the number of transactions flagged.

EDA will help determine whether the selected statistical methods are suitable for the data and whether transformations or robust statistical measures should be considered.

6.5 Statistical Anomaly Detection

Two statistical techniques will be implemented as baseline methods.

6.5.1 Z-Score Analysis

The Z-score measures how many standard deviations an observation lies above or below a feature's mean.

[
z=\frac{x-\mu}{\sigma}
]

Where:

- (x) is the observed feature value.
- (\mu) is the feature mean.
- (\sigma) is the feature standard deviation.

A transaction may be flagged when its absolute Z-score exceeds a threshold, such as 3:

[
|z|>3
]

The initial implementation will calculate feature-level anomaly indicators and combine them into a transaction-level score. For example, the maximum absolute standardized feature value can be used as a simple score.

Because the dataset contains anonymized features and potentially skewed distributions, the threshold of 3 will be treated as a baseline rather than a universally optimal decision rule. Alternative thresholds will be evaluated using the validation data.

6.5.2 Interquartile Range

The IQR identifies observations outside the central range of a feature's distribution.

[
IQR=Q_3-Q_1
]

Where (Q_1) is the first quartile and (Q_3) is the third quartile.

The conventional lower and upper bounds are:

[
L=Q_1-1.5(IQR)
]

[
U=Q_3+1.5(IQR)
]

Observations below (L) or above (U) will be marked as potential outliers.

The IQR method will be applied to selected numerical features, including transaction amount and anonymized components. Feature-level indicators will then be combined into a transaction-level anomaly score.

The analysis will investigate whether a transaction should be flagged when any feature exceeds its bounds or whether a minimum number of unusual features produces a more useful alert rate.

Importantly, an outlier will not automatically be considered fraudulent. The fraud labels will be used only during evaluation to determine how frequently the statistical flags correspond to known fraud cases.

6.6 Machine Learning Anomaly Detection

6.6.1 Isolation Forest

Isolation Forest will be implemented as the primary unsupervised machine learning baseline.

The algorithm isolates observations using randomly selected features and split points. Anomalous observations tend to be isolated in fewer partitioning steps than observations that occur in dense, common regions of the data.

The model will be trained without fraud labels. Its anomaly scores will be compared with the known fraud labels in the validation and test datasets.

The expected advantage over Z-score and IQR is its ability to identify unusual combinations of features rather than relying exclusively on independent feature thresholds.

Model parameters, including the expected anomaly proportion and number of trees, will be selected using validation experiments.

6.6.2 Supervised Random Forest Benchmark

A Random Forest classifier will be trained using the available fraud labels. This model will serve as a supervised benchmark rather than an unsupervised anomaly detector.

The model will be evaluated under the original class distribution. Class weighting will be investigated to reduce the tendency to favour the majority class.

A second experiment may investigate controlled resampling of the training data. All resampling will occur within the training process to prevent synthetic observations from leaking into the validation or test datasets.

The purpose of this benchmark is to determine whether access to labelled fraud examples provides a measurable advantage over methods that learn unusual patterns without labels.

6.7 Deep Learning Anomaly Detection: Autoencoder

A feedforward autoencoder will be implemented as the primary deep learning component.

An autoencoder consists of two main components:

- **Encoder:** Compresses the input transaction features into a lower-dimensional representation.
- **Decoder:** Reconstructs the original input from the compressed representation.

The model will initially be trained on legitimate transactions only. Its objective will be to minimize the reconstruction error between the input features and their reconstructed values.

A transaction-level reconstruction score can be calculated as:

[
E(x)=\frac{1}{d}\sum_{j=1}^{d}(x_j-\hat{x}_j)^2
]

Where:

- (d) is the number of input features.
- (x_j) is the original value of feature (j).

- (\hat{x}_j) is the reconstructed value of feature (j).
- (E(x)) is the reconstruction error for the transaction.

Transactions with high reconstruction errors will be considered potential anomalies. The threshold will be selected using a validation procedure that considers the desired precision-recall trade-off.

The model architecture will use a small number of fully connected layers and a lower-dimensional latent representation. Hyperparameters, such as the latent dimension, learning rate, number of training epochs, and reconstruction threshold, will be tuned using validation data.

The autoencoder is selected because it can learn nonlinear relationships among transaction features while remaining computationally feasible for an academic project.

A generative adversarial network or diffusion model will not be part of the minimum viable implementation. Such models would substantially increase the project's complexity. They may be considered as future extensions if sufficient time and computational resources are available.

6.8 Proposed Hybrid Detection Framework

The main experimental extension will combine statistical anomaly scores with the scores produced by Isolation Forest and the autoencoder.

The proposed hybrid framework will use the following procedure:

1. Calculate the transaction's Z-score-based anomaly score.
2. Calculate the transaction's IQR-based outlier score.
3. Obtain its Isolation Forest anomaly score.
4. Obtain its autoencoder reconstruction error.
5. Normalize the component scores using training or validation data.
6. Combine the scores into a single hybrid anomaly score.
7. Apply a threshold selected using validation data.
8. Compare the resulting flags with the known fraud labels in the test dataset.

A simple weighted combination may be defined as:

[
S(x)=w_zZ(x)+w_iI(x)+w_fF(x)+w_aA(x)
]

Where:

- (Z(x)) is the normalized Z-score anomaly score.
- (I(x)) is the normalized IQR outlier score.
- (F(x)) is the normalized Isolation Forest anomaly score.
- (A(x)) is the normalized autoencoder reconstruction score.
- (w_z,w_i,w_f,w_a) are non-negative weights whose sum equals 1.

The initial experiment will use equal weights. An additional experiment may select weights on validation data, provided that this does not overfit the limited number of fraud cases.

The hybrid approach is intended to test whether different methods identify complementary anomalies. Its effectiveness will be measured rather than assumed.

A separate hybrid experiment may combine the statistical score with the supervised Random Forest output. This will be treated as a distinct configuration because it uses labelled fraud information and therefore has a different training assumption.

The hybrid framework's novelty lies in the systematic evaluation of score-level fusion against its individual components, with attention to class imbalance, false-alert burden, and model complexity. It is an experimental design contribution rather than a claim that score fusion itself is a new algorithm.

6.9 Evaluation Metrics

The models will be evaluated using metrics appropriate for rare-event detection.

Precision

Precision measures the proportion of flagged transactions that are actually fraudulent.

[
Precision=\frac{TP}{TP+FP}
]

Recall

Recall measures the proportion of actual fraudulent transactions correctly identified.

$$
Recall=\frac{TP}{TP+FN}
$$

F1-score

The F1-score is the harmonic mean of precision and recall.

$$
F1=2\frac{Precision\times Recall}{Precision+Recall}
$$

Precision-Recall AUC (PR-AUC)

PR-AUC summarizes the precision-recall trade-off across different thresholds. It is particularly informative when fraudulent transactions are rare.

False-positive rate

$$
FPR=\frac{FP}{FP+TN}
$$

The study will also report the number of false-positive alerts, confusion matrices, ROC-AUC as a secondary metric, and the time required to train and score transactions.

Accuracy will be reported only as a supplementary measure.

For anomaly detection models, PR-AUC will be calculated from continuous anomaly scores rather than only binary predictions. The baseline prevalence of fraud, approximately 0.172%, will be considered when interpreting the results.

6.10 Experimental Comparison

The following models will be compared:

Model	Learning approach	Primary purpose
Z-score	Statistical	Univariate anomaly baseline
IQR	Statistical	Distribution-based outlier baseline
Isolation Forest	Unsupervised ML	Multivariate anomaly detection

Autoencoder	Unsupervised deep learning	Nonlinear reconstruction-based detection
Random Forest	Supervised ML	Label-informed benchmark
Hybrid anomaly score	Combined	Evaluate the value of score fusion

The evaluation will use consistent data splits and an untouched test set. Thresholds and model parameters will be selected using validation data only.

The results will be presented in comparison tables and visualizations, including precision-recall curves, confusion matrices, and comparisons of false-positive alerts at selected operating thresholds.

Where feasible, repeated experiments or bootstrap confidence intervals will be used to assess the stability of the results. The small number of fraudulent examples means that uncertainty must be considered when interpreting differences between models.

7. Tools and Technologies

The project will use the following tools:

- **Python:** Main programming language.
- **Pandas:** Data loading, cleaning, and manipulation.
- **NumPy:** Numerical calculations and statistical operations.
- **Matplotlib and Seaborn:** Exploratory analysis and visualization.
- **Scikit-learn:** Statistical preprocessing, Isolation Forest, Random Forest, and evaluation metrics.
- **TensorFlow/Keras or PyTorch:** Implementation of the autoencoder.
- **Jupyter Notebook:** Experimental workflow, documentation, and results.
- **Kaggle:** Dataset source.

The implementation will prioritize widely available libraries and reproducible experiments.

8. Expected Contributions

The expected contributions are as follows.

8.1 Comparative Evaluation

The project will provide a controlled comparison of simple statistical methods, unsupervised machine learning, supervised classification, and deep learning for fraud detection.

8.2 Hybrid Anomaly Detection

The study will evaluate whether combining different anomaly scores improves detection performance compared with individual models.

8.3 Imbalance-Aware Evaluation

The project will demonstrate why accuracy is insufficient for evaluating fraud detection and investigate the trade-offs between recall, precision, and false-positive alerts.

8.4 Practical Decision Support

The proposed framework will produce anomaly scores that could help fraud analysts prioritize transactions for investigation. The output will be considered a decision-support mechanism, not a replacement for human review or a production banking control.

8.5 Reproducible Implementation

The Python implementation and experimental documentation will provide a foundation for future work involving more advanced methods, additional datasets, and temporal fraud detection.

9. Expected Outcomes and Applications

The project is expected to deliver:

1. An exploratory analysis of a highly imbalanced credit card transaction dataset.
2. Working implementations of Z-score, IQR, Isolation Forest, Random Forest, and autoencoder-based detection.
3. A hybrid anomaly scoring method combining statistical and AI-generated scores.
4. A comparative evaluation using metrics suitable for rare fraud events.
5. An analysis of threshold selection and false-positive alert burden.
6. A discussion of model interpretability, computational cost, and limitations.
7. Recommendations about which methods are most appropriate under different operational priorities.

Potential applications include fraud analyst decision support, transaction screening, exploratory financial data analysis, and the prioritization of suspicious transactions for further investigation.

10. Challenges and Limitations

10.1 Extreme Class Imbalance

Only 492 transactions are fraudulent. This limits the amount of positive-class information available for training and makes the performance estimates sensitive to the selected test cases.

10.2 Statistical Assumptions

Z-score analysis can be affected by skewed distributions and extreme values. IQR is more robust to extreme observations but does not capture complex multivariate relationships when applied independently to features.

10.3 Deep Learning Complexity

Autoencoders require appropriate feature scaling, architecture selection, and threshold calibration. A model trained on legitimate transactions may still reconstruct some fraudulent transactions accurately, while unusual legitimate transactions may produce large reconstruction errors.

10.4 Dataset Representativeness

The dataset covers only two days in 2013 and contains mostly anonymized features. The results may not generalize to contemporary banking systems, different transaction types, or evolving fraud behaviour.

10.5 Evaluation and Information Leakage

Model selection and threshold tuning can produce misleading results if the test set influences training decisions. The project will separate training, validation, and test data and apply any resampling only within the training process.

10.6 Interpretability and Privacy

The anonymized features limit the ability to explain individual transactions in familiar banking terms. Although the project uses public data, data handling, access control, and responsible reporting will follow appropriate research practices.

10.7 Scope Management

Advanced methods such as graph neural networks, GAN-based synthetic fraud generation, and diffusion models may require substantially more implementation effort. They will be treated as potential extensions rather than essential components of the initial project.

11. Project Plan

The project will be completed in the following stages.

Stage	Activities
1. Literature review	Review statistical anomaly detection, fraud detection, class imbalance, and recent AI research
2. Data preparation	Download, inspect, preprocess, and split the dataset
3. Statistical baselines	Implement Z-score and IQR
4. Machine learning	Implement Isolation Forest and Random Forest
5. Deep learning	Develop and validate the autoencoder
6. Hybrid framework	Combine and evaluate anomaly scores
7. Evaluation	Calculate metrics, examine thresholds, and compare results
8. Final reporting	Discuss findings, limitations, and future improvements

The project will prioritize completing and validating the statistical baselines before developing the deep learning and hybrid components. This will ensure that a meaningful comparative study can be completed even if advanced modelling requires additional time.

12. Conclusion

This proposal presents a quantitative study of statistical anomaly detection for fraudulent credit card transactions. It retains Z-score and IQR as interpretable baselines while extending the investigation to Isolation Forest, a supervised Random Forest benchmark, and an autoencoder-based deep learning model.

The central research question is whether the additional complexity of machine learning and deep learning produces measurable improvements over statistical methods when the dataset is extremely imbalanced. A hybrid anomaly scoring framework will further

investigate whether combining complementary approaches can improve fraud detection while maintaining a manageable number of false alerts.

By using appropriate evaluation metrics, consistent experimental procedures, and recent research as a foundation, the project aims to produce a practical and reproducible assessment of anomaly detection techniques. The findings will help clarify the circumstances in which simple statistical methods are sufficient and when more advanced AI techniques may offer additional value.

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